

第 4 章 ~~quiz~~

$$1. \lim_{n \rightarrow \infty} (n^3 \sin \frac{2}{n} - 2n^2) = \lim_{n \rightarrow \infty} (n^3 [\frac{2}{n} - \frac{1}{3!}(\frac{2}{n})^3 + \dots] - 2n^2) = -\frac{2^3}{3!} = -\frac{4}{3}$$

证法: let $t = \frac{1}{n}$

$$\lim_{t \rightarrow 0^+} \frac{\sin(2t) - 2t}{t^3} = \lim_{t \rightarrow 0^+} \frac{[2t - \frac{1}{3!}(2t)^3 + \dots] - 2t}{t^3} = -\frac{4}{3}$$

$$2. \lim_{n \rightarrow \infty} ((n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1})$$

$$= \lim_{n \rightarrow \infty} [(n^2 - n)(1 + \frac{1}{n} + \frac{1}{2!n^2} + \dots) - n^2 \sqrt{1 + \frac{1}{n^4}}]$$

$$= \lim_{n \rightarrow \infty} [(\cancel{n^2} + \frac{1}{2!} - \frac{1}{2!n} + \frac{1}{3!n^2} + \dots) - \cancel{n^2}(1 + \frac{1}{2n^4} + \dots)]$$

$$= -\frac{1}{2}$$

$$3. \lim_{x \rightarrow 0} \frac{\cos(\sin x) - \sqrt{1-x^2}}{x^a} = \lim_{x \rightarrow 0} \frac{(\cos(\sin x) - 1) + (1 - \sqrt{1-x^2})}{x^a} = \lim_{x \rightarrow 0} \frac{(-\frac{1}{2}(\sin x)^2 + \frac{1}{2}x^2)}{x^a} = \lim_{x \rightarrow 0} \frac{-\frac{1}{2}(\frac{x^2}{2!} + \dots) + \frac{1}{2}x^2}{x^a} = b$$

$$\Rightarrow a = 4$$

$$b = \frac{1}{6} + \frac{1}{8} = \frac{4+3}{24} = \frac{7}{24}$$

$$4. (1) \lim_{x \rightarrow 1} \frac{x^2 - 1}{\ln x} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{\ln(1+(x-1))} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}} = 2$$

$$(2) \lim_{x \rightarrow \infty} (\sqrt[6]{x^6 + x^5} - \sqrt[6]{x^6 - x^5}) = \lim_{x \rightarrow \infty} x (\sqrt[6]{1 + \frac{1}{x}} - \sqrt[6]{1 - \frac{1}{x}}) = \lim_{x \rightarrow \infty} x [(1 + \frac{1}{6x} + \dots) - (1 - \frac{1}{6x} + \dots)] = \frac{1}{3}$$

$$(3) \lim_{x \rightarrow \infty} x^2 (\sqrt{1 + \frac{1}{x}} + (1 - \frac{1}{x})^{\frac{1}{2}} - 2) = \lim_{x \rightarrow \infty} x^2 [(1 + \frac{1}{2x} - \frac{1}{8x^2} + \dots) + (1 - \frac{1}{2x} - \frac{1}{8x^2} + \dots) - 2] = -\frac{1}{4}$$

$$(4) \lim_{x \rightarrow \infty} [x - x^2 \ln(1 + \frac{1}{x})] = \lim_{x \rightarrow \infty} [x - x^2 (\frac{1}{x} - \frac{1}{2x^2} + \dots)] = \frac{1}{2}$$

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参考点: $\rightarrow$

$$1. f(x) = \int_0^x \frac{1}{1-t} dt = \int_0^x \sum_{n=0}^{\infty} t^n dt = \sum_{n=0}^{\infty} \int_0^x t^n dt = \sum_{n=0}^{\infty} \left[\frac{t^{n+1}}{n+1} \right]_0^x = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}, |x| < 1.$$

$$2. f(x) = \ln(x + \sqrt{x^2 + 1}) \Rightarrow f'(x) = \frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}} = (1 + x^2)^{-\frac{1}{2}} = 1 - \frac{x^2}{2} + \left(\frac{1}{6}\right)x^4 + \dots$$

$$\Rightarrow f(x) = x - \frac{x^3}{6} + \frac{1}{24}\left(\frac{1}{6}\right)x^5 + \dots$$

$$3. f(x) = \frac{1}{(2-x)^2} = \left(\frac{1}{2-x}\right)' = \left(\frac{1}{2} \cdot \frac{1}{1-\frac{x}{2}}\right)' = \left(\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n\right)' = \left(\sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}\right)' = \sum_{n=0}^{\infty} \frac{nx^{n-1}}{2^{n+1}} = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{2^{n+1}} = \sum_{n=0}^{\infty} \frac{(n+1)x^n}{2^{n+2}} = \sum_{n=0}^{\infty} \frac{(n+1)}{2^{n+2}} x^n, -2 < x < 2$$

$$4. f(x) = \tan^{-1}\left(\frac{1+x}{1-x}\right) \Rightarrow f'(x) = \frac{1 + \frac{1+x}{1-x}}{1 + \left(\frac{1+x}{1-x}\right)^2} = \frac{2}{2(1+x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$\Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} + \frac{\pi}{4}, -1 \leq x \leq 1$$

$$5. f(x) = \frac{4}{x^2 - 2x + 5} + \ln x = \frac{4}{(x-1)^2 + 4} + \ln[(1+x-1)] = \frac{4}{4} \cdot \frac{1}{1 + \frac{(x-1)^2}{4}} + \ln[1+(x-1)]$$

$$\text{参考点: } = \sum_{n=0}^{\infty} (-1)^n \left(\frac{(x-1)^2}{4}\right)^n + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n+1} (x-1)^{n+1}, |x-1| < 1$$

$$1. f(x) = \sum_{n=1}^{\infty} \frac{n+1}{n(n+1)} x^n = \sum_{n=1}^{\infty} \left[\frac{1}{n+1} + \frac{1}{n} \left(\frac{1}{n} - \frac{1}{n+1} \right) \right] x^n = 2 \sum_{n=1}^{\infty} \frac{1}{n} x^n - \sum_{n=1}^{\infty} \frac{x^n}{n+1}$$

$$= 2 \sum_{n=1}^{\infty} \frac{x^n}{n} - \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = 2 \int_0^x \sum_{n=1}^{\infty} \frac{1}{n} t^{n-1} dt - \frac{1}{x} \int_0^x \sum_{n=1}^{\infty} t^n dt = 2 \int_0^x \frac{1}{1-t} dt - \frac{1}{x} \int_0^x \frac{t}{1-t} dt$$

$$= 2 \ln(1-x) - \frac{1}{x} \int_0^x \left(\frac{1}{1-t} - 1 \right) dt = 2 \ln(1-x) + \frac{\ln(1-x)}{x} + 1, |x| < 1, x \neq 0.$$

$$\text{if } x=0, f(x)=0.$$

$$2. \sum_{n=0}^{\infty} \frac{n^2+1}{2^n \cdot n!} x^n = \sum_{n=0}^{\infty} \frac{n}{(n-1)!} \cdot \frac{x^n}{2^n} + \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{x}{2}\right)^n = x \left(\sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)! 2^n} \right)' + e^{\frac{x}{2}}$$

$$= x \left(\frac{x}{2} e^{\frac{x}{2}} \right)' + e^{\frac{x}{2}} = x \left(\frac{1}{2} e^{\frac{x}{2}} + \frac{x}{2} e^{\frac{x}{2}} \right) + e^{\frac{x}{2}} = \left(\frac{x^2}{2} + \frac{x}{2} + 1 \right) e^{\frac{x}{2}}$$